Mathematics 1
Assignment 1
Title: Derivatives (Lec.1+Lec. 2+ Lec. 3)

Textbook: Calculus, Early Transcendental Functions, Fourth Edition, by: Robert T. Smith, Roland B. Minton, McRraw-Hill Publications.

(a) Find the first derivative of the following functions

(1) $f(x) = Ln(sin^{-1} x)$

(2) $f(x) = 3^{x^2+1}(4x^5 - 6)$

(3) $f(x) = (1 + x^3)\, cosh(\, ln\, x)$

(4) $f(x) = \frac{tan(3x+1)}{x^5-1}$

(5) $f(x) = Ln(Lnx)$

(6) $f(x) = \frac{1+Lnx}{sin\ \ (2x-1)}$

(7) $y = cosh^{-1}(\, x^4 + 1)$

(8) $y = e^{4x+1}\, tanh^{-1}(\, x + 2)$

(9) $y = tan^{-1}(\, sin\, x)$

(10) $y = (3x^2 - 4x + 6)\text{sech}^{-1}\, x$

(11) $y^2 + \tan xy - e^x - y - x^3$

(12) $y = x^{\sin^{-1} x}$

(13) $y = x^{lnx} + e^{3x}$

(b) if $y = \sec^{-1} x$, Show that $\acute{y} = \frac{1}{x\sqrt{x^2-1}}$.